

Mohamed Shameer Khan A

Chennai | +91 7200855958 | shameerkhanmohamed@gmail.com

<https://www.linkedin.com/in/mohamed-shameer-khan-a-27175431b> <https://github.com/Moshakn111>

Summary

Data Science and Machine Learning practitioner with a strong focus on Computer Vision and intelligent systems. Experienced in developing practical applications and working with data preprocessing, model development, and analysis. Skilled in object detection, data visualization, and building scalable machine learning solutions.

Experience

AI Engineer Intern – Phosphene AI

- Optimized large scale computer vision datasets and pipelines using Python, OpenCV, and CVAT, improving data processing efficiency by **30%** and ensuring high quality annotations.
- Developed an **unsupervised behavior-based approach** for traffic flow analysis to detect anomalies, improving detection precision by **15%**.
- Enhanced traffic violation detection systems using motion-based logic and temporal consistency filtering for improved robustness in real world scenarios.

Projects

Smart Traffic Violation Detection

- Built a real-time computer vision system using YOLOv8 and SORT for vehicle detection and tracking, achieving **25% faster inference** through optimized processing.
- Expanded an **unsupervised behavior based approach** to detect red light and wrong lane violations from traffic flow patterns, improving precision by **18%** without relying on signal recognition.
- Reduced false positives by **20%** using temporal consistency and direction based analysis, and leveraged **Lightning AI SaaS Platform** for efficient model training and experimentation.

Welding Defect Classification using Laser Scanner Data (Ongoing)

- Designed a pipeline to convert 1D laser scanned weld signals into 2D representations (GAF/MTF) for effective deep learning-based defect classification.
- Benchmarking multiple models including CNNs (VGG16, ResNet, MobileNet, InceptionNet), Transformers (ViT), and Mamba to identify the most accurate and robust approach.
- Improving robustness to noise and variability in weld data, while integrating **Grad-CAM (XAI)** for visual explanations and automated report generation.

Business Intelligence Dashboards (Power BI & Tableau)

- Developed interactive dashboards using Power BI and Tableau, integrating multi source data to deliver reliable KPI tracking and enable data driven decision making.
- Built dynamic visualizations and automated reporting workflows to simplify analysis and provide actionable insights for stakeholders.

Skills

Programming: Python, R, SQL, Java, C, HTML/CSS

Frameworks/Tools: TensorFlow, Scikit-learn, Power BI, Tableau, Gemini, Claude, CVAT, Roboflow, Kaggle, VS Code, Anaconda Navigator, Google Colab, GitHub, Hugging Face, Lightning AI, Flask

Core Areas: Data Science, Machine Learning, Deep learning, Data Visualization, Computer Vision, Statistics

Soft Skills: Problem Solving, Leadership, Adaptability, Teamwork, Time Management, Work Ethic, Persistence

Certifications

- OCI Data Science Professional – Oracle
- Python for Data Science – NPTEL
- Business Analytics using R (part 2) – NPTEL
- Power BI Data Analyst Associate – Microsoft